LU Solver on the GPU

From FP64 kernels to tensor cores, watched by NixNan

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SC26 tutorial

Algorithm: LU with partial pivoting, solve $Ax = b$

$$PA = LU, \qquad \underbrace{p = \arg\max_{i \geq k} |U_{ik}|}_{\text{pivot}}, \qquad \underbrace{L_{jk} = \frac{U_{jk}}{U_{kk}}}_{\text{multiplier}}, \qquad \underbrace{U_{jl} \mathrel{-}= L_{jk} U_{kl}}_{\text{eliminate, } j>k,\ l \geq k}$$

$$Ly = Pb \ \text{(forward)}, \qquad Ux = y \ \text{(backward)}$$

